BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

## Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Breaks a task into smaller problems.

Q2. A state unable to reach any valid solution.

Q3. To stop further calls at a terminal state.

Q4. Popped 5; bottom-to-top [2,8].

### You Do pages · Independent questions

Q1. 2+3+4=9 [2] overshoots [1], undo3 [1], 2+4=6 [2].

Q2. Check sum==target success [2], end list failure [1], choose branch recurse [1], undo/exclude branch recurse [1], combine outcomes [1].

Q3. Negative later values can reduce sum [2]; e.g. target4, start6 then −2 [2]; pruning at6 loses valid solution [2].

### Exit Ticket · Model answer

False. Negative later numbers could lower it.

## Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**





BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

# LESSON 068 · Lesson 068 · Backtracking: Choices, Dead Ends and Undo

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What does divide and conquer do?

Q2. What is a dead end in a search?

Q3. Why must recursion have a base case?

Q4. Stack Push 2, Push 5, Pop, then Push 8: give popped item and remaining stack.

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

## I Do · Read new knowledge

### Key vocabulary

backtracking — Try and undo choices.

search tree — Branches of possible decisions.

pruning — Stop provably impossible branch.

state — Current partial solution data.

### New learning

Define and predict — Backtracking explores a choice, tests whether its state remains viable, then undoes the choice if it cannot lead to a solution.

Explain — A search tree records choices and alternative branches. Pruning avoids expanding branches already proven invalid.

Evaluate — A correct search needs a stopping condition. Pruning on sum>target only works when remaining numbers cannot reduce the sum (e.g. non-negative inputs).

### Worked example

Find a subset of [1,3,4] with total 5.

1. Follow the reasoning — Backtrack after overshoot; alternative [1,4] reaches the target.

## We Do · Practise with guidance

Find a subset of [2,5,7] with target 9.

```
Choose2,5 →7; add7 →14 reject; undo5; choose7 →9.
```

Follow the reasoning

The alternative branch 2+7 gives a valid solution.





BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Trace first failed and one successful branch for subset sum target6 from [2,3,4]. [6]

### Question 2

Explain a recursive backtracking design with choose/exclude and two stopping conditions. [6]

### Question 3

Explain why pruning when sum>target may be invalid if negative integers occur; give counterexample. [6]

## Exit Ticket · Answer before checking

A branch is always impossible when its subtotal exceeds the target, even if negatives remain.